

MACHINE LEARNING FOR SIGNAL PROCESSING

10-2-2025

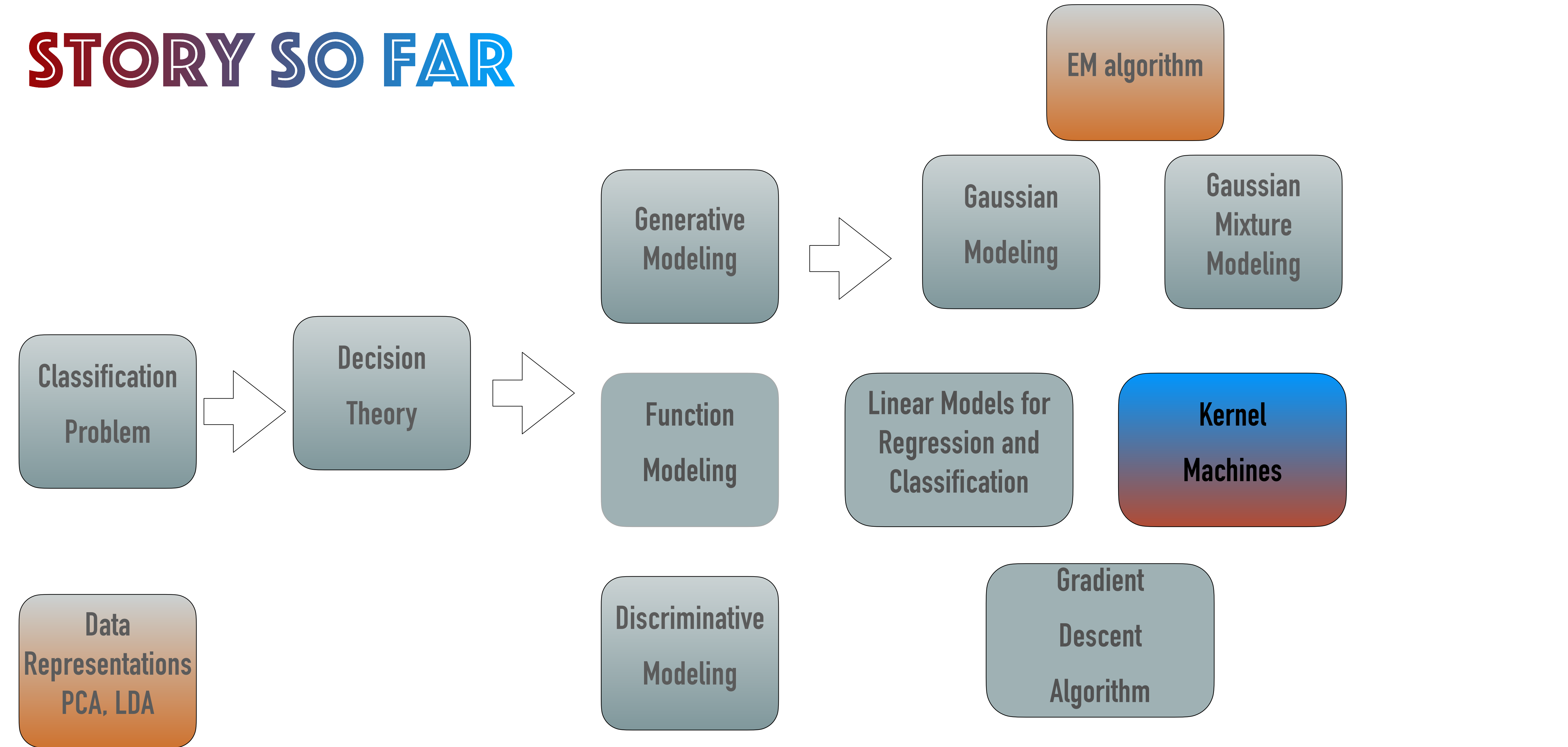


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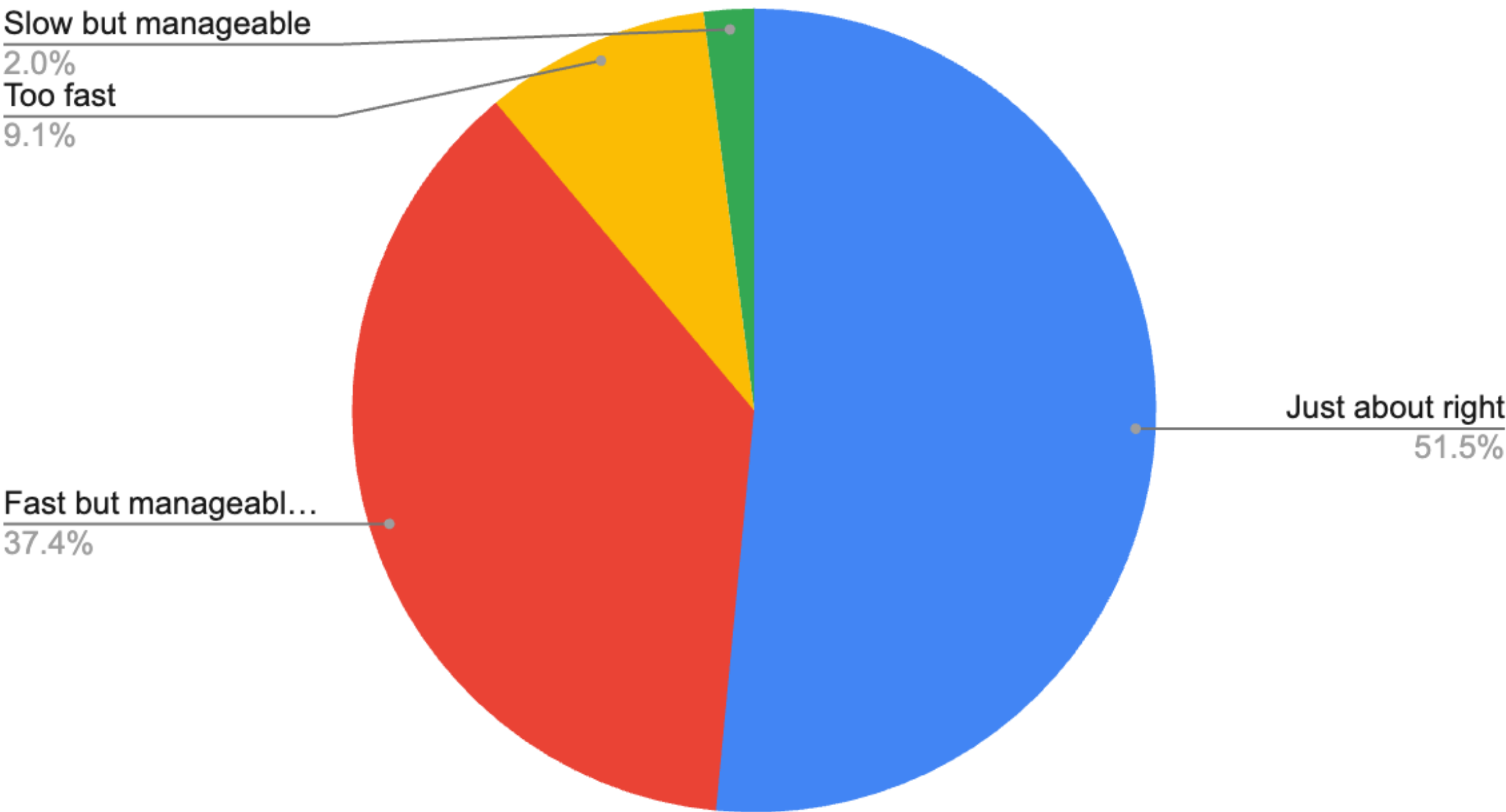
<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>

STORY SO FAR

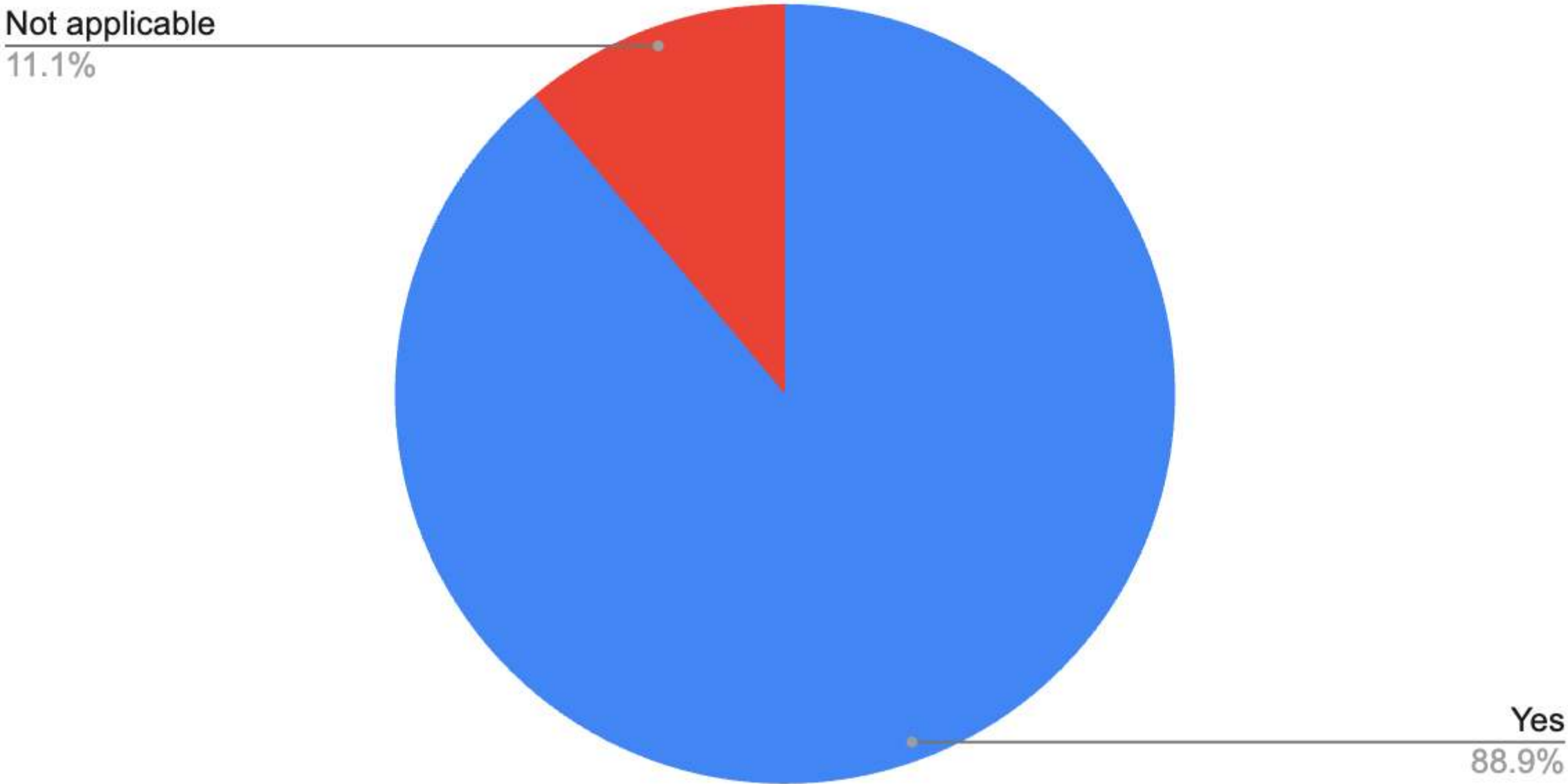


FEEDBACK ABOUT YOUR FEEDBACK

Count of How do you find the pace of the course?

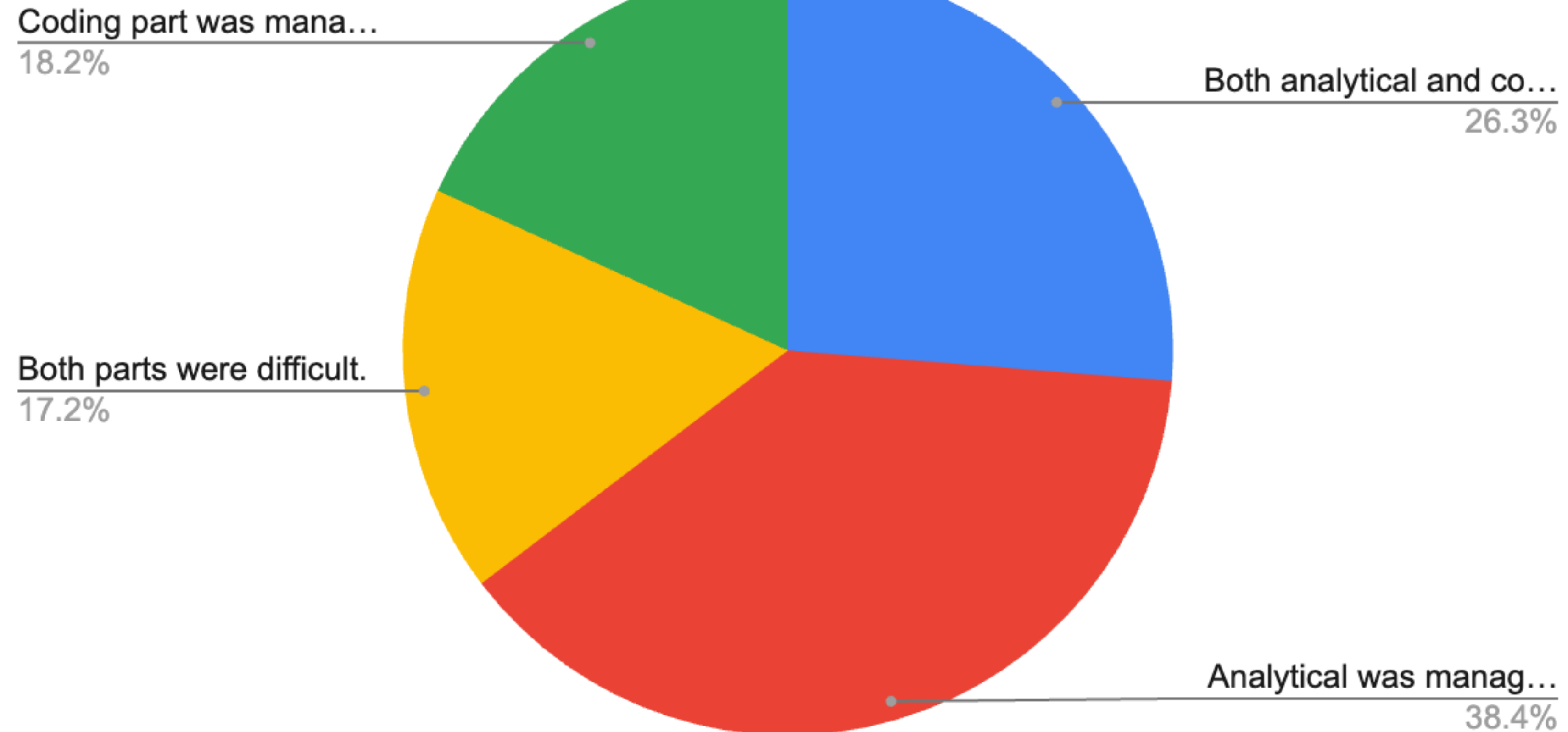


Count of Do you think your questions were valued if you raise them during the class ?



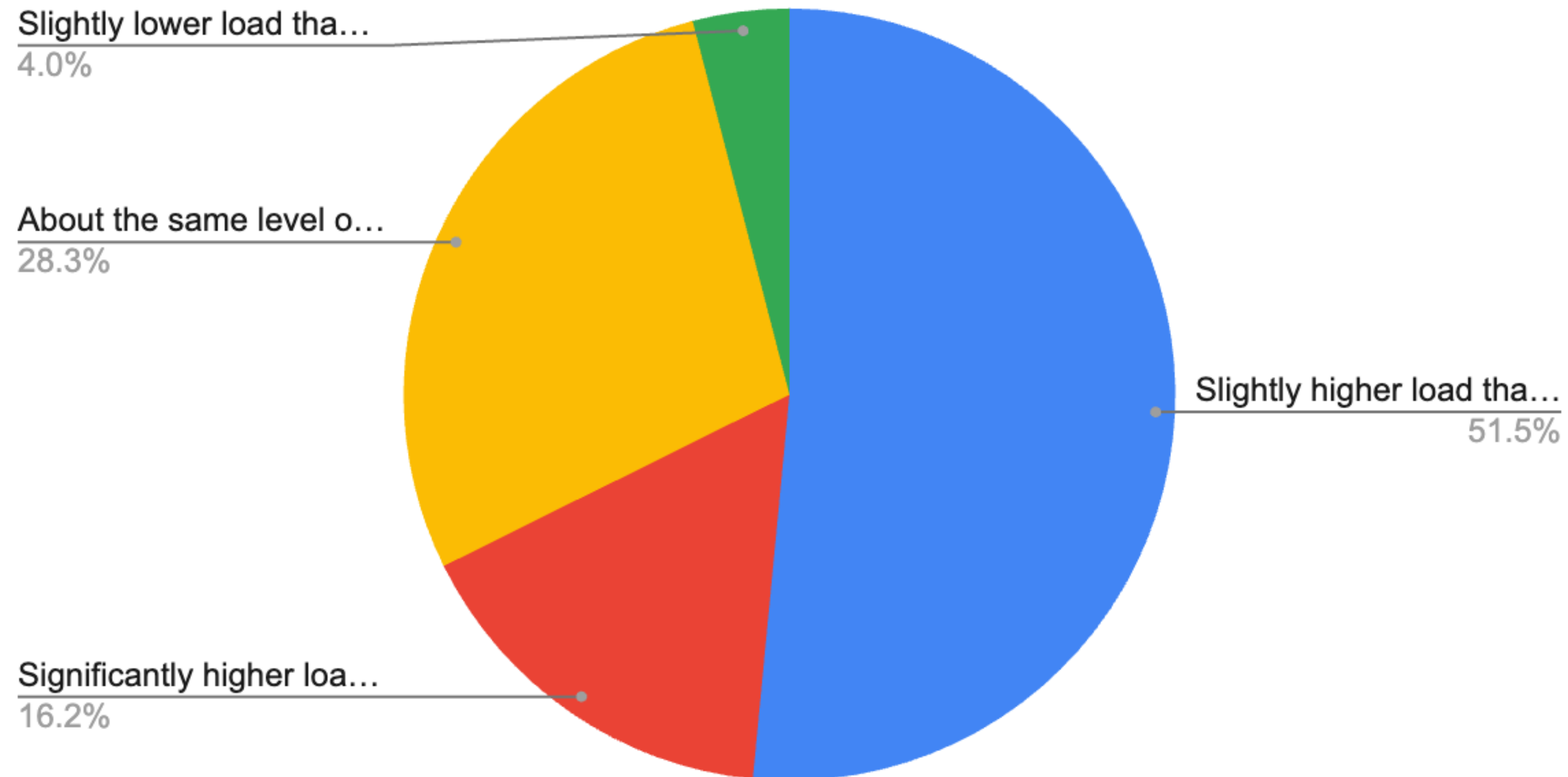
FEEDBACK ABOUT YOUR FEEDBACK

Count of How involved was the assignment (both analytical and coding)



FEEDBACK ABOUT YOUR FEEDBACK

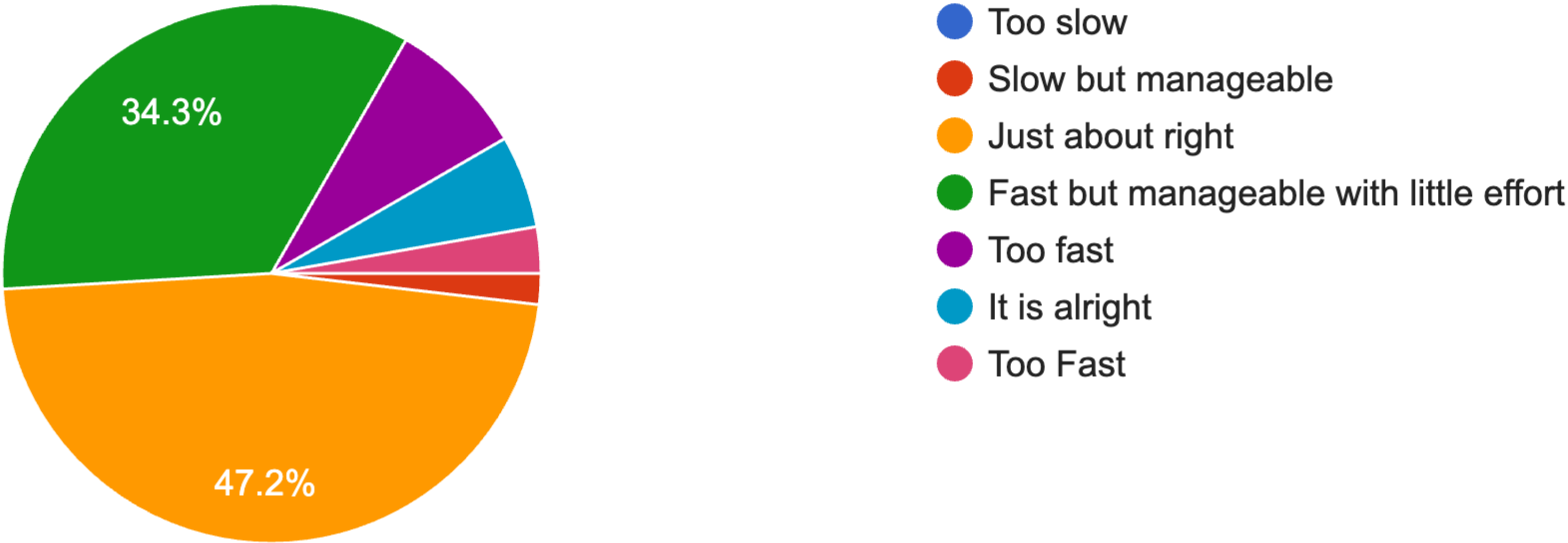
Count of How do you rate the overall load of the course w.r.t. other courses you have taken at IISc so far.



FEEDBACK ABOUT YOUR FEEDBACK

How do you find the pace of the course?

108 responses



COMMENTS - GOOD, BAD, UGLY

The course is being taught extremely well. The only problem I had till now was in coding assignment the data for Question 6 was hard to deal with. I was able to perform similar operation on randomly generated data of same dimension but with data given it took too much time.

If possible, please try to elaborate more with some examples in simple way in the class as me and many of us are not able to understand the concepts.

I think some derivation the Instructor leaves for homework are doable, some are not. So, if it can be done in the class, it'd be helpful.

Some questions felt a bit ambiguous in the coding ones, like what are the features and which are the data points, I would be grateful if those things are more explicitly defined, or a wrapper code is provided.

While starting a new topic, please start with why, where and how that topic is useful so that the reader can stick with the problem and understand it's solution methodology.

After the assignments' deadline, if we receive the solutions to the questions asked in the assignment, we would get valuable feedback for improving in the remainder of the course.

I want to get some practice questions for midterm.

Some analytical relevant practice questions with solution manual will help students to develop skills in problem solving and developing understanding of the subject in a better way. Some Tutorials for coding will also help students in understanding the concepts taught in the class.

Mathematical derivation part can be a bit slower. Sometimes it becomes difficult to catch up with so many equations and appreciate what each of them is doing.

I think we need to work hard enough to manage the pace, difficulty level and everything.

The most difficult part I feel about this course is, it requires me to read a lot of extra material like bishop to understand the fabric connecting all the topics taught in the class.

COMMENTS - GOOD, BAD, UGLY

1. More intuition of how to see it as a system rather than a black box and understand each step of this system.
2. A min or two recap of some of the necessary concepts for the present one being taught.
3. Connecting the links of some of the concepts taught from this subject and other subjects to understand, where and when it is used.
4. Some history about how and why the particular concept was discovered or needed.
5. I'm really happy with the way the classes are going melodic and would also the journey to continue the same 😊🙏.

The professor's teaching style is excellent, truly one of the best classes here at IISc. After the lectures, almost all doubts are cleared. The lectures are well-structured and very clearly explained.

I have only one issue with the course, which is related to the assignment deadlines. The questions in Assignment 1 were not particularly hard, but given that almost all courses in the even semester are assignment- and project-based, having a little more than 10 days to complete an assignment would be preferable. Perhaps each assignment could have an even 2-week deadline? The challenge of time management doesn't lie in the analytical part but in the coding. We're learning to use so many tools in different courses this semester that it's hard to keep track of how each one works when juggling multiple assignments. For example, understanding how Librosa works took quite a bit of time, figuring out what the different parameters do, what `mel_spectrogram` is, etc.

A little more time for assignments would be very helpful. I understand that the same argument could be made even if we had 14 days, and some people probably still would. However, considering that upcoming assignments in each course will likely become even more challenging and time-consuming, I kindly request you to consider this.

COMMENTS - GOOD, BAD, UGLY

There should be some practice question sets before the Mid-term exams.

The course progression is good, especially the part where you go back to the flowchart to explain the bigger picture about the models, it helps us keep context of what we are learning.

It would very helpful if the mathematical insights (such as properties of positive semidefinite matrices for example) that we encounter during the lectures were either revised or even just plainly stated in the class notes.

There should be some practice question sets before the Mid-term exams.

The course progression is good, especially the part where you go back to the flowchart to explain the bigger picture about the models, it helps us keep context of what we are learning.

Able to understand concepts well from the class itself as the pace is not too fast.

The best Machine learning course I found in IISC in terms of teaching .I don't know how much marks I will get , but I am enjoying it.

The course is really good, every class focuses on an important topic, coherence is maintained, subtle points of each topic is discussed in each class.
Thank you

Excellent class, specially story so far section in classroom. I would like to take more course offered by Sriram Sir.

Sir, you are really gem, while taking this course i was not sure that it will meet expectations but after few session this you proved that course is really nice structured and you are also gem.

ANNOUNCEMENTS

- ❖ Midterm exam
 - ✓ Feb 15 [Morning 10am, B308, B306, B313]
 - Instructions (same as those in the practice Questions distributed)
 - ➔ Open book etc ...
- ❖ Attendance form

HOW DO WE FIND THE MODEL PARAMETERS

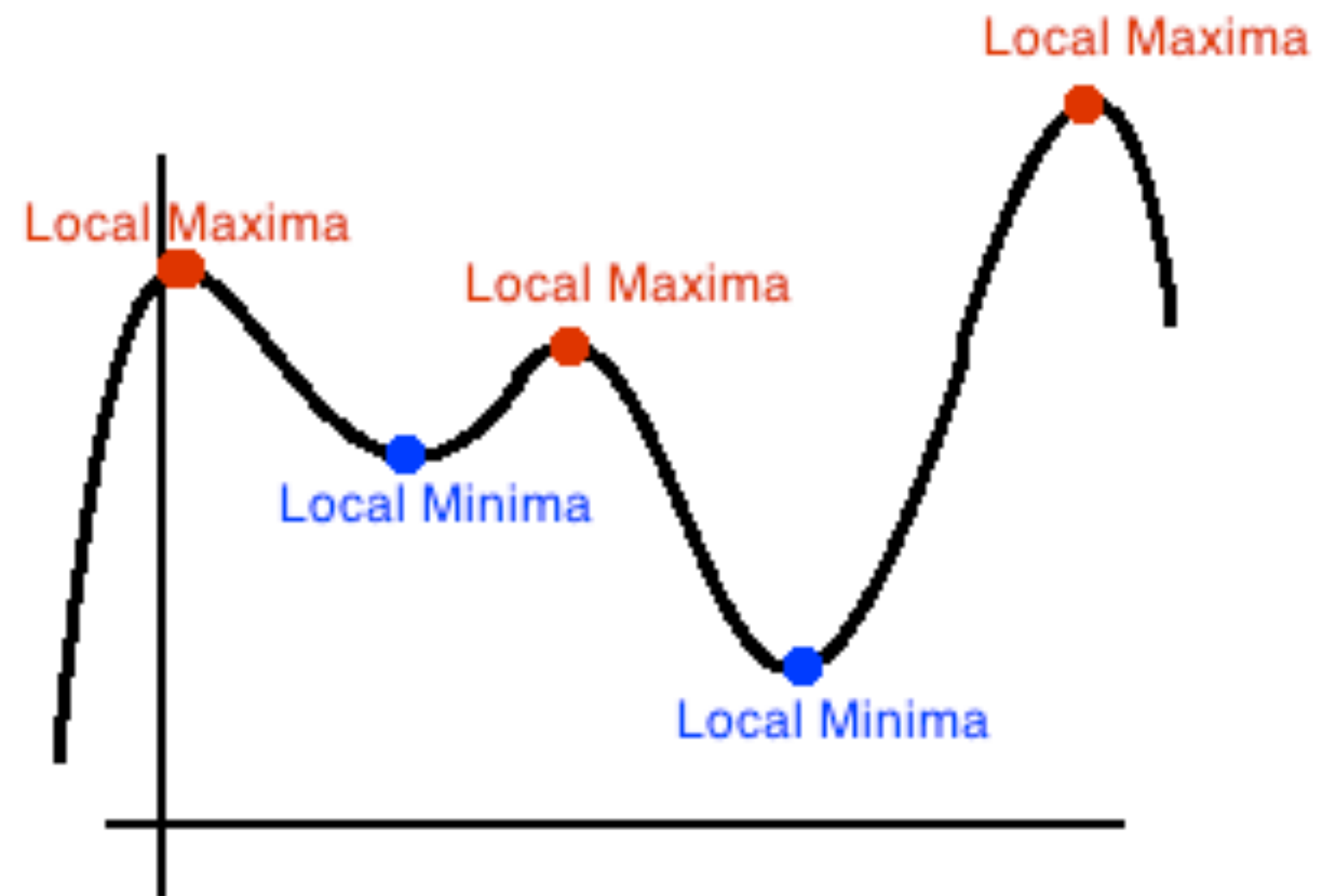
- ❖ We want to maximize the total probability
 - Or equivalently minimize the negative log probability

$$E(\mathbf{w}, b) = - \left[\sum_{n=1}^N t_n \log y_n + (1 - t_n) \log (1 - y_n) \right]$$
$$y_n = \sigma(\mathbf{w}^T \mathbf{x}_n + b)$$

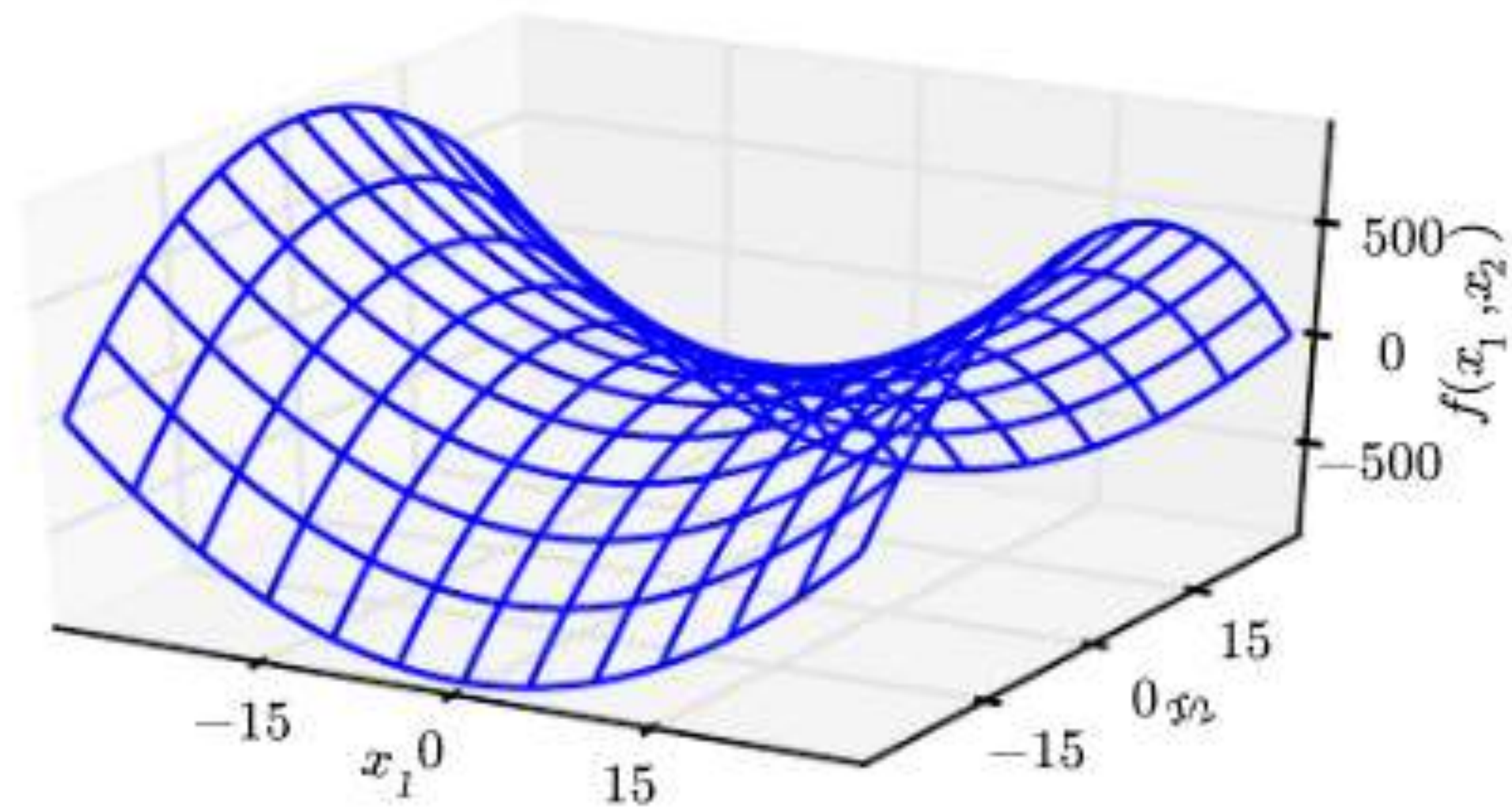
- The parameters of the model need to be solved based on minimizing the error
 - This error function is non-convex in the parameters
- Need to perform iterative update —> Gradient descent

NON-CONVEX OPTIMIZATION

- ❖ Local maxima and minima

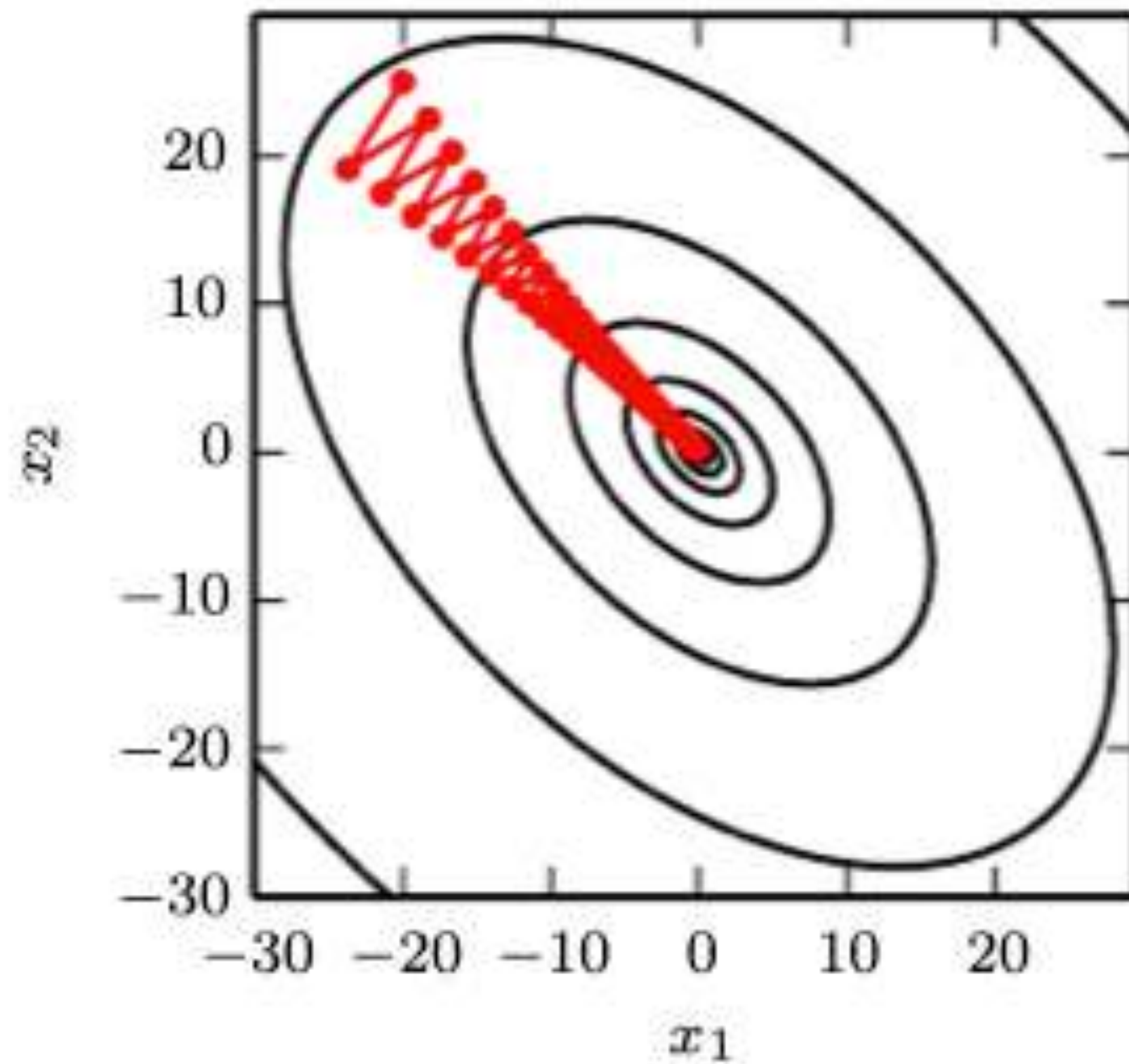


GRADIENT DESCENT



Multi-dimensional saddle points

GRADIENT DESCENT



Slowness of gradient descent
For a quadratic function

GRADIENT DESCENT ALGORITHM

- ❖ Start with random weights

$$\text{iter } t = 0 \rightarrow \mathbf{w}^0, b^0$$

- ❖ Compute gradients and update the model iteratively

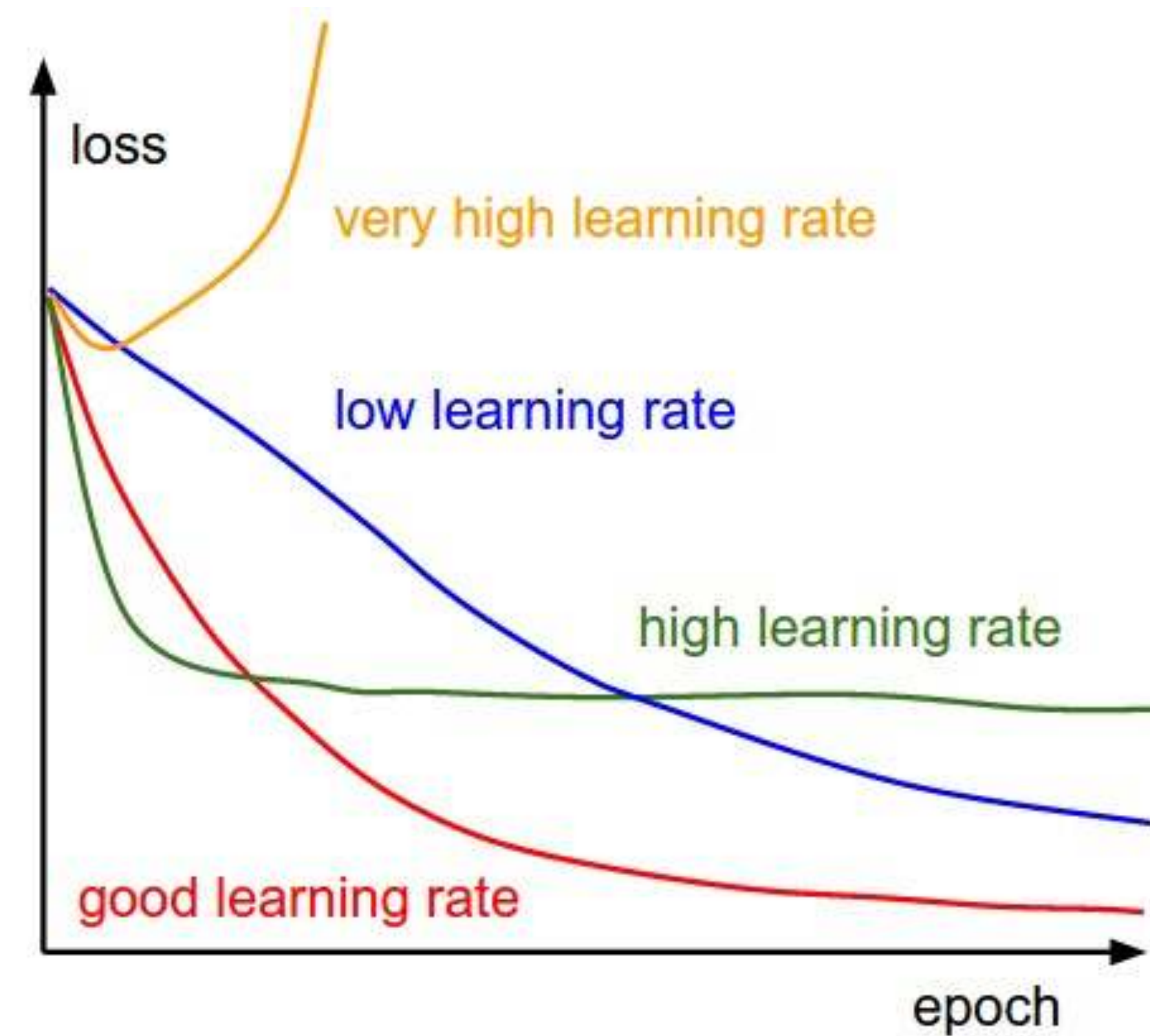
$$\mathbf{w}^t = \mathbf{w}^{t-1} - \eta \frac{\partial E}{\partial \mathbf{w}} \bigg|_{\mathbf{w}, b = \mathbf{w}^{t-1}, b^{t-1}}$$

$$b^t = b^{t-1} - \eta \frac{\partial E}{\partial b} \bigg|_{\mathbf{w}, b = \mathbf{w}^{t-1}, b^{t-1}}$$

❖ Check for convergence $\rightarrow$ If not, $t = t + 1$ and go to the step above.

LEARNING RATE

- Solving a non-convex optimization.
- Iterative solution.
- Depends on the initialization.
- Convergence to a local optima.



MULTI-CLASS LOGISTIC REGRESSION

❖ Outputs are

$$\mathbf{y} = \text{softmax}(\mathbf{W}^T \mathbf{x} + \mathbf{b})$$

❖ Targets are one hot encoded vectors

❖ Error function

$$E(\mathbf{W}, \mathbf{b}) = - \sum_n \sum_k t_{nk} \log y_{nk}$$

$$\mathbf{t} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ 0 \\ 1 \\ 0 \\ \cdot \\ 0 \end{bmatrix}$$

➤ cross entropy function

as before, learnt using gradient descent

STOCHASTIC GRADIENT DESCENT

- ❖ The total error is a sum of errors in all the data samples

$$E(\mathbf{w}, b) = - \left[\sum_{n=1}^N t_n \log y_n + (1 - t_n) \log (1 - y_n) \right]$$

- May be too slow - requiring a large number of updates before convergence.
- Can we take a subset of the data and perform gradient descent and update the model on the subset ?

STOCHASTIC GRADIENT DESCENT

- ❖ Split the data into random chunks



- perform gradient updates on each mini batch
 - assumption - gradients computed on the mini-batch well approximate the full batch gradients

STOCHASTIC GRADIENT DESCENT

- ❖ Split the data into random chunks



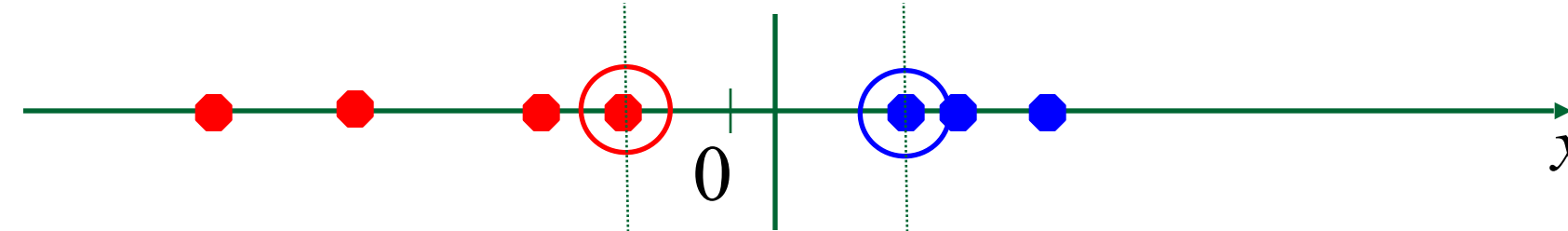
- hyper-parameters

LINEAR REGRESSION REVISITED

- ❖ Primal and dual forms
- ❖ Solution in dual space
- ❖ Kernels

KERNEL MACHINES

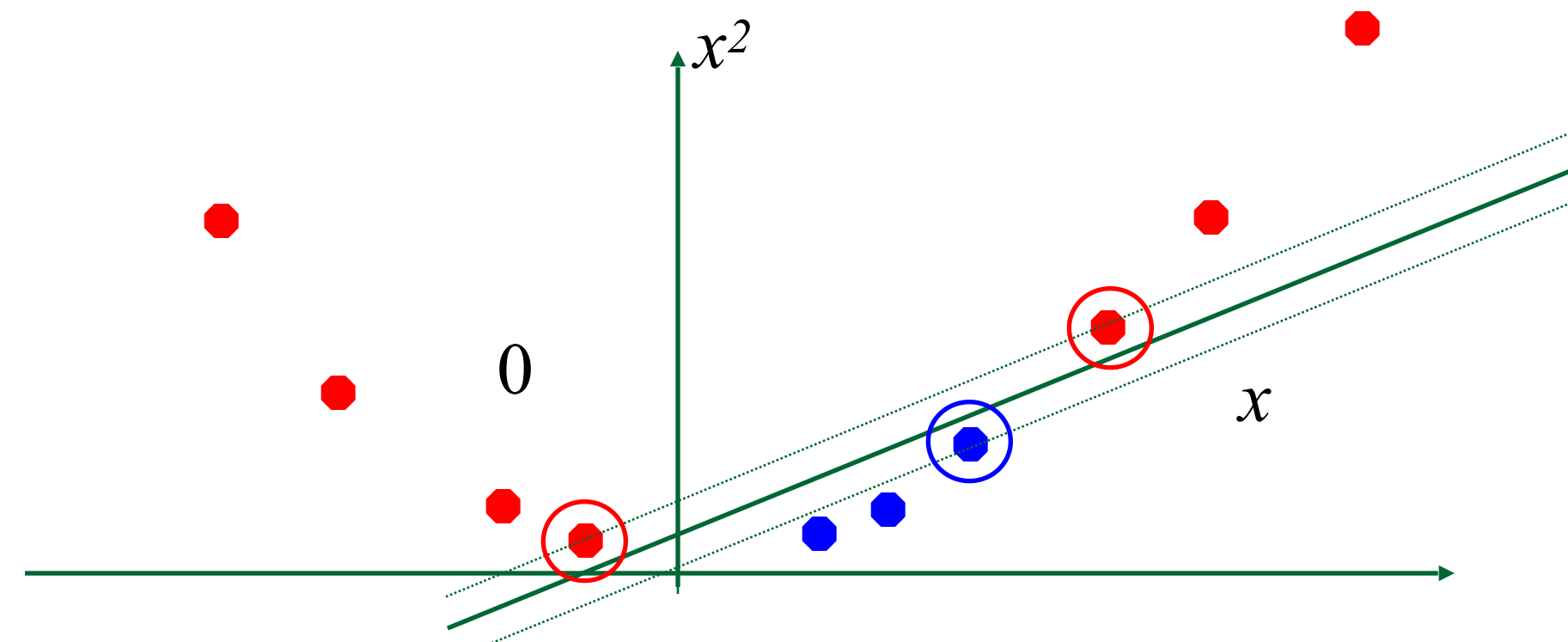
- Datasets that are linearly separable with some noise work out great:



- But what are we going to do if the dataset is just too hard?

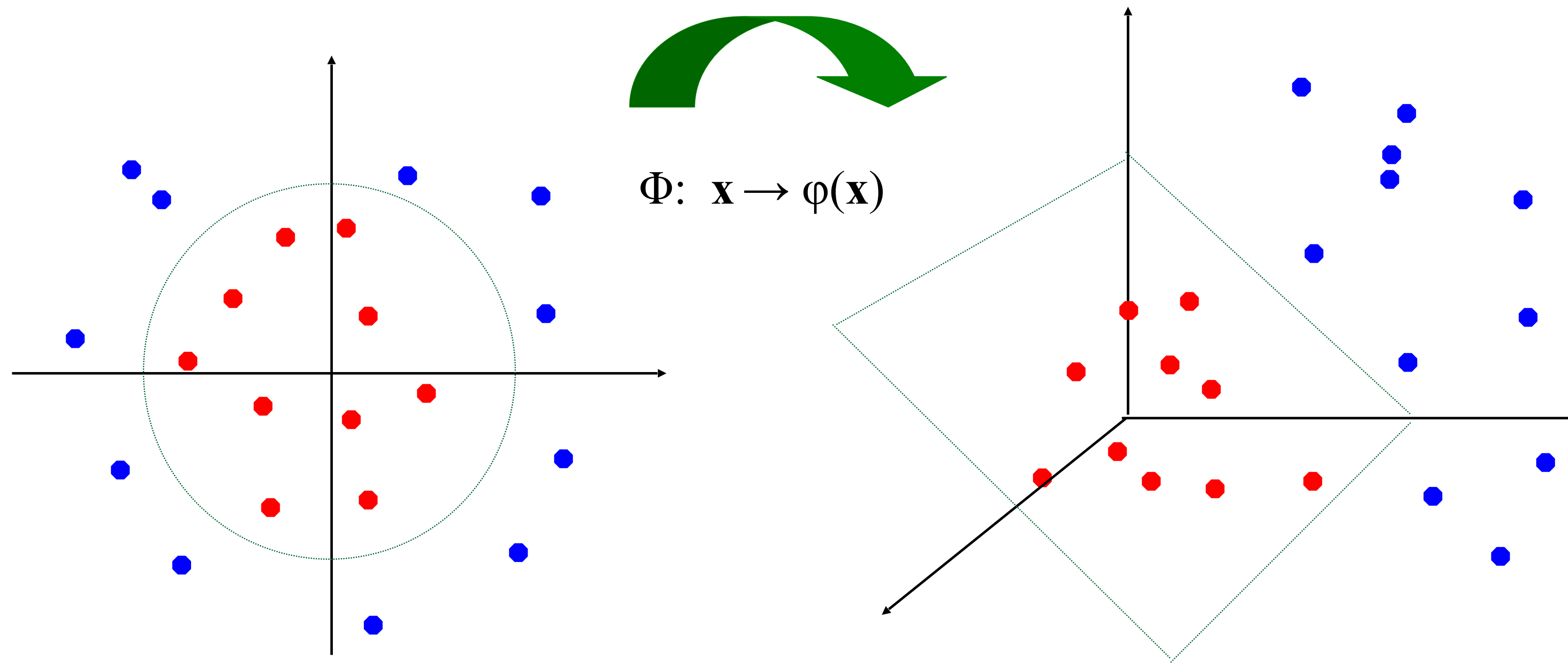


- How about... mapping data to a higher-dimensional space:



KERNEL TRICK

- General idea: the original input space can always be mapped to some higher-dimensional feature space where the training set is separable:



The “Kernel Trick”

- The linear classifier relies on dot product between vectors $k(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$
- If every data point is mapped into high-dimensional space via some transformation $\Phi: \mathbf{x} \rightarrow \phi(\mathbf{x})$, the dot product becomes:

$$k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

- A *kernel function* is some function that corresponds to an inner product in some expanded feature space.
- Example:

2-dimensional vectors $\mathbf{x} = [x_1 \ x_2]$; let $k(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^2$,

Need to show that $K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$:

$$\begin{aligned} k(\mathbf{x}_i, \mathbf{x}_j) &= (1 + \mathbf{x}_i^T \mathbf{x}_j)^2, \\ &= 1 + x_{i1}^2 x_{j1}^2 + 2 x_{i1} x_{j1} x_{i2} x_{j2} + x_{i2}^2 x_{j2}^2 + 2 x_{i1} x_{j1} + 2 x_{i2} x_{j2} \\ &= [1 \ x_{i1}^2 \ \sqrt{2} x_{i1} x_{i2} \ x_{i2}^2 \ \sqrt{2} x_{i1} \ \sqrt{2} x_{i2}]^T [1 \ x_{j1}^2 \ \sqrt{2} x_{j1} x_{j2} \ x_{j2}^2 \ \sqrt{2} x_{j1} \ \sqrt{2} x_{j2}] \\ &= \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j), \quad \text{where } \phi(\mathbf{x}) = [1 \ x_1^2 \ \sqrt{2} x_1 x_2 \ x_2^2 \ \sqrt{2} x_1 \ \sqrt{2} x_2] \end{aligned}$$

KERNELS

- For many functions $k(\mathbf{x}_i, \mathbf{x}_j)$ checking that $k(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$ can be cumbersome.
- Mercer's theorem: Every semi-positive definite symmetric function is a kernel
 - Semi-positive definite symmetric functions correspond to a semi-positive definite symmetric Gram matrix:

K =

$k(\mathbf{x}_1, \mathbf{x}_1)$	$k(\mathbf{x}_1, \mathbf{x}_2)$	$k(\mathbf{x}_1, \mathbf{x}_3)$	...	$k(\mathbf{x}_1, \mathbf{x}_N)$
$k(\mathbf{x}_2, \mathbf{x}_1)$	$k(\mathbf{x}_2, \mathbf{x}_2)$	$k(\mathbf{x}_2, \mathbf{x}_3)$		$k(\mathbf{x}_2, \mathbf{x}_N)$
...	...	...	...	...
$k(\mathbf{x}_N, \mathbf{x}_1)$	$k(\mathbf{x}_N, \mathbf{x}_2)$	$k(\mathbf{x}_N, \mathbf{x}_3)$	...	$k(\mathbf{x}_N, \mathbf{x}_N)$

EXAMPLES OF KERNEL FUNCTIONS

- Linear: $k(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^T \mathbf{x}_j$
- Polynomial of power p : $k(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^T \mathbf{x}_j)^p$
- Gaussian (radial-basis function network):

$$k(\mathbf{x}_i, \mathbf{x}_j) = \exp \frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\sigma^2}$$

- Sigmoid: $k(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\beta_0 \mathbf{x}_i^T \mathbf{x}_j + \beta_1)$

PROPERTIES OF KERNEL FUNCTIONS

$$k(\mathbf{x}, \mathbf{x}') = ck_1(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = f(\mathbf{x})k_1(\mathbf{x}, \mathbf{x}')f(\mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = q(k_1(\mathbf{x}, \mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = \exp(k_1(\mathbf{x}, \mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}') + k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = k_1(\mathbf{x}, \mathbf{x}')k_2(\mathbf{x}, \mathbf{x}')$$

$$k(\mathbf{x}, \mathbf{x}') = k_3(\phi(\mathbf{x}), \phi(\mathbf{x}'))$$

$$k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{A} \mathbf{x}'$$

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a) + k_b(\mathbf{x}_b, \mathbf{x}'_b)$$

$$k(\mathbf{x}, \mathbf{x}') = k_a(\mathbf{x}_a, \mathbf{x}'_a)k_b(\mathbf{x}_b, \mathbf{x}'_b)$$

THANK YOU

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